Needs-based recommendation systems

Machine Learning Pipeline for Financial Needs Prediction and Product Recommendation

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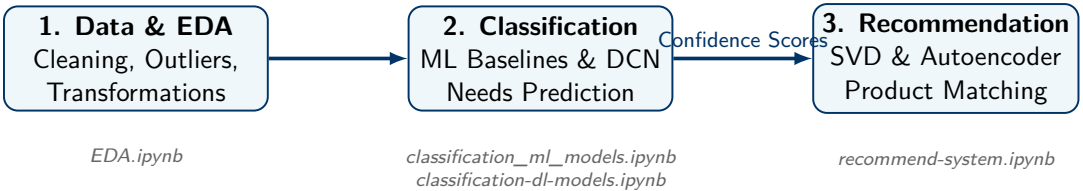
Project Overview



Dataset Structure

Data Quality Report

Key EDA Findings

Feature Transformation and Engineering

1. Data Transformation (`DataTransformer`)

- **Numeric Conversion:** Coercion of mixed/string formats to continuous variables.
- **Log Transformation:** Applied $\log(1 + x)$ to highly skewed financial variables (`Wealth`, `Income`) to normalize distributions.
- **Scaling:** `StandardScaler` applied post-engineering for stable convergence.

2. Household Metrics

- `WealthIncomeRatio`
- `IncomePerFamilyMember`

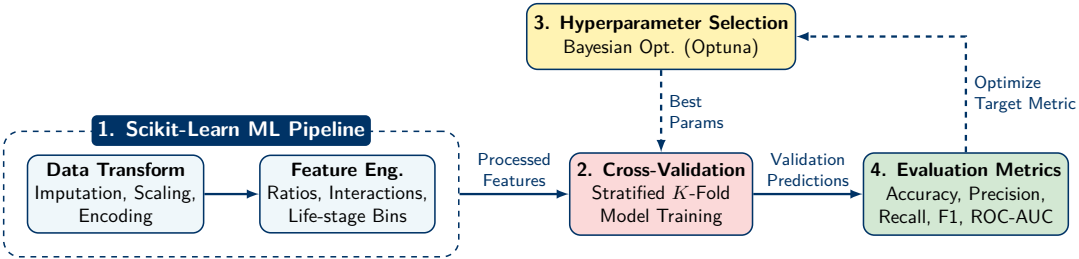
3. Interaction Features

- `RiskEducationInteraction` ($Risk \times Edu$)
- `RiskWealthInteraction` ($Risk \times \log(Wealth)$)
- `AgeRiskInteraction` ($Age \times Risk$)

4. Non-linearity & Life-Stages

- `AgeSquared`: Models the "Life-Cycle" curve (wealth peaking in mid-life).
- **Age Bins**: One-hot flags for `Under35`, `35_54`, `55_69`, and `70plus`.

ML Pipeline Overview



Baseline Models Evaluated

Baseline Results: AccumulationInvestment

Baseline Results: IncomeInvestment

Optuna Hyperparameter Tuning

Optimization Results

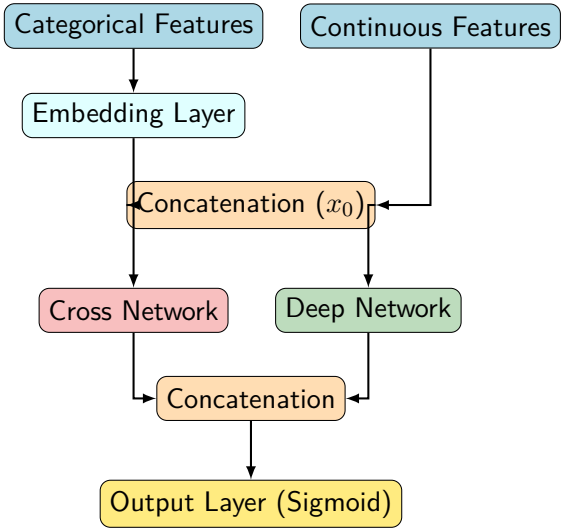
XAI Methods: SHAP, LIME & Permutation

Income Investment Model Insights

Accumulation Investment Model Insights

LIME: Local Interpretability Example

Deep Learning Model Architecture: DCN [1]



Multi-Task Learning Results

Task	Acc	Prec	Rec	F1	AUC
Income	0.783	0.852	0.544	0.664	0.767
Accum.	0.795	0.846	0.708	0.771	0.852

Architecture Highlights

- **Cross Network**: Applies explicit bounded-degree feature crossing efficiently.
- **Deep Network**: Learns implicit non-linear higher-order interactions.
- **Joint Training**: Both networks share the same concatenated input embeddings, optimising for dual investment targets.

Methodology: User-Item Matrix

We construct a user-item interaction matrix R using predicted probabilities from the supervised classifiers, penalised by risk misalignment.

Matrix Entry R_{ij}

$$R_{ij} = P_{\text{type}(j)}(i) \cdot \big(1 - \alpha \cdot |\text{RiskPropensity}(i) - \text{Risk}(j)|\big)$$

- $P_{\text{type}(j)}(i)$: Predicted probability of client i needing product j's type (Accumulation or Income).
- $\alpha \in [0, 1]$: Decay factor penalising risk misalignment.
  - **Economic Rationale**: Reflects the firm's risk aversion and regulatory suitability constraints (e.g., MiFID II).
  - **Trade-off**: Balances maximising expected fee revenue/conversion against minimising compliance risks and long-term customer churn from unsuitable investments.

Methodology: Models & Evaluation

We use model-based CF [2] leveraging matrix R:

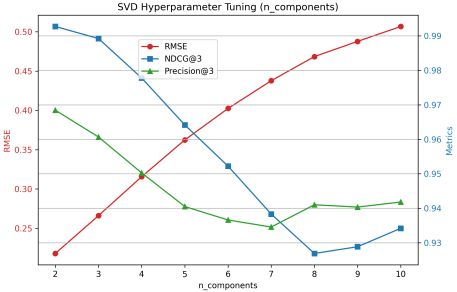
- **Truncated SVD**: Factorises $R \approx U_k \Sigma_k V_k^T$ into low-dimensional latent factors to capture hidden preferences.
- **Autoencoder**: Learns non-linear representations by compressing and reconstructing the interaction matrix ($z_u = f_\theta(r_u)$, $\hat{r}_u = g_\phi(z_u)$).

Evaluation Metrics

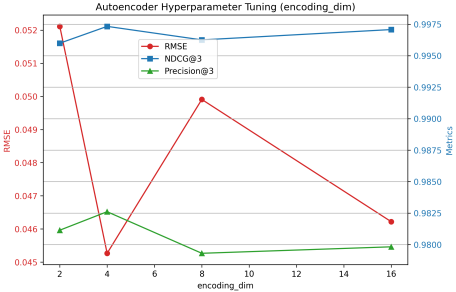
- **RMSE**: Measures the model's reconstruction error.
- **Precision@k & Recall@k**: Evaluate the relevance of the top-k recommendations.
- **NDCG@k**: Assesses ranking quality, prioritising relevant items placed higher.

Experimental Results

Model	RMSE	Precision@3	Recall@3	NDCG@3
Truncated SVD	0.402	0.937	0.937	0.952
Autoencoder	**0.039**	**0.984**	**0.984**	**0.997**



SVD Tuning



Autoencoder Tuning

Project Summary

References

[1] Weiyu Guo et al. "Explainable enterprise credit rating using deep feature crossing". en. In: *Expert Systems with Applications* 220 (June 2023), p. 119704.

[2] Marwa Sharaf et al. "A survey on recommendation systems for financial services". In: *Multimedia Tools and Applications* 81.12 (May 2022), pp. 16761–16781.